

Foundations of Anomaly detection

[Anomaly detection]

1.0 Content Block

Explainable anomaly detection Many of the methods discussed above only yield an anomaly score prediction, which often can be explained to users as the point being in a region of low data density (or relatively low density compared to the neighbor's densities). In explainable artificial intelligence, the users demand methods with higher explainability. Some methods allow for more detailed explanations:

2.0 Content Block

Definition Many attempts have been made in the statistical and computer science communities to define an anomaly. The most prevalent ones include the following, and can be categorised into three groups: those that are ambiguous, those that are specific to a method with pre-defined thresholds usually chosen empirically, and those that are formally defined:

3.0 Content Block

Specific Let T be observations from a univariate Gaussian distribution and O a point from T . Then the z-score for O is greater than a pre-selected threshold if and only if O is an outlier.

4.0 Content Block

Density Density-based techniques (k-nearest neighbor, local outlier factor, isolation forests, and many more variations of this concept) Subspace-base (SOD), correlation-based (COP) and tensor-based outlier detection for high-dimensional data One-class support vector machines (OCSVM, SVDD)

5.0 Content Block

Methods Many anomaly detection techniques have been proposed in literature. The performance of methods usually depend on the data sets. For example, some may be suited to detecting local outliers, while others global, and methods have little systematic advantages over another when compared across many data sets. Almost all algorithms also require the setting of non-intuitive parameters critical for performance, and usually unknown before application. Some of the popular techniques are mentioned below and are broken down into categories:

6.0 Content Block

Preprocessing Preprocessing data to remove anomalies can be an important step in data analysis, and is done for a number of reasons. Statistics such as the mean and standard deviation are more accurate after the removal of anomalies, and the visualisation of data can also be improved. In supervised learning, removing the anomalous data from the dataset often results in a statistically significant increase in accuracy.

7.0 Content Block

Petroleum industry Anomaly detection is crucial in the petroleum industry for monitoring critical machinery. Martí et al. used a novel segmentation algorithm to analyze sensor data for real-time anomaly detection. This approach helps promptly identify and address any irregularities in sensor readings, ensuring the reliability and safety of petroleum operations.

8.0 Content Block

Neural networks Replicator neural networks, autoencoders, variational autoencoders, long short-term memory neural networks Bayesian networks Hidden Markov models (HMMs) Minimum Covariance Determinant Deep Learning Convolutional Neural Networks (CNNs): CNNs have shown exceptional performance in the unsupervised learning domain for anomaly detection, especially in image and video data analysis. Their ability to automatically and hierarchically learn spatial hierarchies of features from low to high-level patterns makes them particularly suited for detecting visual anomalies. For instance, CNNs can be trained on image datasets to identify atypical patterns indicative of defects or out-of-norm conditions in industrial quality control scenarios. Simple Recurrent Units (SRUs): In time-series data, SRUs, a type of recurrent neural network, have been effectively used for anomaly detection by capturing temporal dependencies and sequence anomalies. Unlike traditional RNNs, SRUs are designed to be faster and more parallelizable, offering a better fit for real-time anomaly detection in complex systems such as dynamic financial markets or predictive maintenance in machinery, where identifying temporal irregularities promptly is crucial. Foundation models: Since the advent of large-scale foundation models that have been used successfully on most downstream tasks, they have also been adapted for use in anomaly detection and segmentation. Methods utilizing pretrained foundation models include using the alignment of image and text embeddings (CLIP, etc.) for anomaly localization, while others may use the inpainting ability of generative image models for reconstruction-error based anomaly detection.

9.0 Content Block

IT infrastructure In IT infrastructure management, anomaly detection is crucial for ensuring the smooth operation and reliability of services. These are complex systems, composed of many interactive elements and large data quantities, requiring methods to process and reduce this data into a human and machine interpretable format. Techniques like the IT Infrastructure Library (ITIL) and monitoring frameworks are employed to track and manage system performance and user experience. Detected anomalies can help identify and pre-empt potential performance degradations or system failures, thus maintaining productivity and business process effectiveness.

10.0 Content Block

Parameter-free Also referred to as frequency-based or counting-based, the simplest non-parametric anomaly detection method is to build a histogram with the training data or a set of known normal instances, and if a test point does not fall in any of the histogram bins mark it as anomalous, or assign an anomaly score to test data based on the height of the bin it falls in. The size of bins are key to the effectiveness of this technique but must be determined by the implementer. A more sophisticated technique uses kernel functions to approximate the distribution of the normal data. Instances in low probability areas of the distribution are then considered anomalies.

11.0 Content Block

The Subspace Outlier Degree (SOD) identifies attributes where a sample is normal, and attributes in which the sample deviates from the expected. Correlation Outlier Probabilities (COP) compute an error vector of how a sample point deviates from an expected location, which can be interpreted as a counterfactual explanation: the sample would be normal if it were moved to that location.

12.0 Content Block

In data analysis, anomaly detection (also referred to as outlier detection and sometimes as novelty detection) is generally understood to be the identification of rare items, events or observations which deviate significantly from the majority of the data and do not conform to a well defined notion of normal behavior. Such examples may arouse suspicions of being generated by a different mechanism, or appear inconsistent with the remainder of that set of data. Anomaly detection finds application in many domains including cybersecurity, medicine, machine vision, statistics, neuroscience, law enforcement and financial fraud to name only a few. Anomalies were initially searched for clear rejection or omission from the data to

aid statistical analysis, for example to compute the mean or standard deviation. They were also removed to better predictions from models such as linear regression, and more recently their removal aids the performance of machine learning algorithms. However, in many applications anomalies themselves are of interest and are the observations most desirous in the entire data set, which need to be identified and separated from noise or irrelevant outliers. Three broad categories of anomaly detection techniques exist. Supervised anomaly detection techniques require a data set that has been labeled as "normal" and "abnormal" and involves training a classifier. However, this approach is rarely used in anomaly detection due to the general unavailability of labelled data and the inherent unbalanced nature of the classes. Semi-supervised anomaly detection techniques assume that some portion of the data is labelled. This may be any combination of the normal or anomalous data, but more often than not, the techniques construct a model representing normal behavior from a given normal training data set, and then test the likelihood of a test instance to be generated by the model. Unsupervised anomaly detection techniques assume the data is unlabelled and are by far the most commonly used due to their wider and relevant application.